

Companion XXVIII

The Integrated Sachs–Wolfe Effect and CMB–LSS Cross-Correlations in the Relaxation-Driven Cyclic Cosmology

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Abstract

The Integrated Sachs–Wolfe (ISW) effect and the cross-correlation between the cosmic microwave background (CMB) and large-scale structure (LSS) provide a direct probe of the time evolution of gravitational potentials. In the standard cosmological model, the ISW signal arises from the decay of potentials during the late-time accelerated expansion. However, current measurements show mild tensions with Λ CDM predictions, including excess large-scale correlations and hints of scale-dependent anomalies.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified growth of structure, the suppression of small-scale power, and the enhanced coherence of large-scale potentials alter the time evolution of the gravitational potential. RDCC predicts smoother potentials, reduced small-scale decay, enhanced large-scale coherence, and a characteristic turnover scale tracing the relaxation scale k_{rel} .

This Companion develops the RDCC predictions for the ISW effect, the CMB temperature–galaxy cross-correlation, the CMB lensing–ISW bispectrum, and the scale-dependent potential evolution. We show that RDCC produces distinctive signatures in the ISW auto- and cross-spectra, and we provide forecasts for Planck, ACT, SPT, Euclid, LSST, and SKA. These results establish the ISW effect as a precision probe of the relaxation dynamics of RDCC.

1 Motivation and Overview

The Integrated Sachs–Wolfe (ISW) effect and the cross-correlation between the cosmic microwave background (CMB) and large-scale structure (LSS) provide a direct probe of the time evolution of gravitational potentials. Unlike galaxy clustering or gravitational lensing, which measure the spatial distribution of matter, the ISW effect is sensitive to the temporal evolution of the potential wells and hills that photons traverse on their way from the last scattering surface.

In the standard cosmological model, the ISW effect arises from the decay of gravitational potentials during the late-time accelerated expansion. However, current measurements show mild but persistent tensions with Λ CDM predictions, including excess large-scale correlations, hints of scale-dependent anomalies, and discrepancies between different tracers of large-scale structure.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified growth of structure, the suppression of small-scale power, and the enhanced coherence of large-scale potentials alter the time evolution of the gravitational potential. RDCC predicts smoother poten-

tials, reduced small-scale decay, enhanced large-scale coherence, and a characteristic turnover scale tracing the relaxation scale k_{rel} .

This Companion develops the RDCC predictions for the ISW effect, the CMB temperature–galaxy cross-correlation, the CMB lensing–ISW bispectrum, and the scale-dependent potential evolution. We show that RDCC produces distinctive signatures in the ISW auto- and cross-spectra, and we provide forecasts for Planck, ACT, SPT, Euclid, LSST, and SKA.

1.1 Observational Context

The ISW effect is generated when CMB photons traverse evolving gravitational potentials. This produces temperature anisotropies of the form

$$\frac{\Delta T}{T} = 2 \int_{\eta_{\text{ls}}}^{\eta_0} d\eta \dot{\Phi}(\eta, \mathbf{x}), \quad (1)$$

where $\dot{\Phi}$ is the time derivative of the gravitational potential.

Observationally, the ISW effect is detected through:

- the large-scale CMB temperature anisotropies,
- the CMB–galaxy cross-correlation,
- the CMB–lensing–ISW bispectrum,
- tomographic cross-correlations with LSS tracers.

Current measurements show:

- a mild excess in the CMB–LSS cross-correlation amplitude,
- hints of scale-dependent deviations from Λ CDM,
- inconsistencies between different galaxy samples,
- possible anomalies at low multipoles.

These features motivate exploring alternative models of potential evolution.

1.2 RDCC Perspective

In the Relaxation-Driven Cyclic Cosmology, the relaxation rate $\Gamma(a)$ modifies the growth of structure and the evolution of gravitational potentials. As shown in Companion XVI–XIX, RDCC predicts:

- suppressed small-scale matter power,
- smoother gravitational potentials,
- enhanced large-scale coherence,
- reduced small-scale potential decay,
- a characteristic turnover at k_{rel} .

These effects directly influence the ISW signal:

- **Large scales** ($k < k_{\text{rel}}$): enhanced ISW due to coherent potential evolution.
- **Intermediate scales** ($k \sim k_{\text{rel}}$): characteristic turnover in $\dot{\Phi}$.
- **Small scales** ($k > k_{\text{rel}}$): suppressed ISW due to smoother potentials.

Thus, the ISW effect provides a direct probe of the relaxation scale k_{rel} .

1.3 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC predictions for the ISW effect,
- to compute the RDCC potential evolution and $\dot{\Phi}(k, a)$,
- to analyze the RDCC CMB–galaxy cross-correlation,
- to derive the RDCC ISW–lensing bispectrum,
- and to compare RDCC predictions with Planck, ACT, SPT, Euclid, LSST, and SKA.

These results build directly on the RDCC structure formation framework developed in Companion XVI–XIX and the observational modules of Companion XXV–XXVII.

1.4 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC potential evolution and ISW kernel.
- Section 3 computes the RDCC CMB–galaxy cross-correlation.
- Section 4 analyzes the RDCC ISW–lensing bispectrum.
- Section 5 develops the RDCC predictions for large-scale anomalies.
- Section 6 compares RDCC predictions with Planck, ACT, SPT, Euclid, LSST, and SKA.

Together, these results establish the ISW effect as a precision probe of the relaxation dynamics of RDCC.

2 RDCC Potential Evolution and the ISW Kernel

The Integrated Sachs–Wolfe (ISW) effect is generated by the time evolution of gravitational potentials along the line of sight of CMB photons. In the standard cosmological model, the late-time ISW effect arises from the decay of potentials during the accelerated expansion. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified growth of structure, the suppression of small-scale power, and the enhanced coherence of large-scale potentials alter the time evolution of the potential and therefore the ISW signal.

This section derives the RDCC potential evolution, the RDCC ISW kernel, and the scale-dependent potential decay rate.

2.1 Gravitational Potential Evolution

In linear theory, the gravitational potential is

$$\Phi(k, a) = -\frac{3}{2} \frac{\Omega_m H_0^2}{k^2} \frac{\delta(k, a)}{a}. \quad (2)$$

The time derivative is

$$\dot{\Phi}(k, a) = -\frac{3}{2} \frac{\Omega_m H_0^2}{k^2} \left[\frac{\dot{\delta}(k, a)}{a} - \frac{\delta(k, a)\dot{a}}{a^2} \right]. \quad (3)$$

Using $\dot{\delta} = H f \delta$ yields

$$\dot{\Phi}(k, a) = -\frac{3}{2} \frac{\Omega_m H_0^2}{k^2} \frac{H \delta(k, a)}{a} [f(a) - 1]. \quad (4)$$

Thus, potential decay is controlled by the deviation of the growth rate from unity.

2.2 RDCC Growth Rate and Potential Evolution

In RDCC, the growth rate becomes scale-dependent:

$$f_{\text{RDCC}}(k, a) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

Thus, the RDCC potential evolution is

$$\dot{\Phi}_{\text{RDCC}}(k, a) = \dot{\Phi}(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

RDCC predicts:

- suppressed small-scale potential decay,
- enhanced large-scale coherent evolution,
- a turnover at k_{rel} ,
- smoother potential time derivatives.

2.3 RDCC Potential Coherence

The potential coherence length is

$$\lambda_{\text{coh}} \sim k_{\text{rel}}^{-1}. \quad (7)$$

RDCC predicts:

- larger coherence length,
- reduced small-scale potential fluctuations,
- enhanced large-scale potential alignment.

These effects directly influence the ISW signal.

2.4 ISW Kernel

The ISW temperature anisotropy is

$$\frac{\Delta T}{T} = 2 \int_{\eta_s}^{\eta_0} d\eta \dot{\Phi}(k, \eta). \quad (8)$$

The ISW kernel is defined as

$$W_{\text{ISW}}(k, a) = 2 \dot{\Phi}(k, a). \quad (9)$$

In RDCC:

$$W_{\text{ISW, RDCC}}(k, a) = W_{\text{ISW}}(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (10)$$

This predicts:

- enhanced ISW at large scales,
- suppressed ISW at small scales,
- a turnover at k_{rel} ,
- smoother ISW kernels.

2.5 RDCC ISW Power Spectrum

The ISW auto-spectrum is

$$C_\ell^{\text{ISW}} = 4\pi \int \frac{dk}{k} \mathcal{P}_\Phi(k) \left| \int d\eta \dot{\Phi}(k, \eta) j_\ell(k\chi) \right|^2. \quad (11)$$

In RDCC:

$$C_{\ell, \text{RDCC}}^{\text{ISW}} = C_\ell^{\text{ISW}} \left[1 - \chi_\ell \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right], \quad (12)$$

where

$$\ell_{\text{rel}} \sim k_{\text{rel}} \chi(z). \quad (13)$$

This predicts:

- enhanced low- ℓ ISW power,
- suppressed high- ℓ ISW power,
- a turnover at ℓ_{rel} ,
- smoother ISW spectra.

2.6 Summary

RDCC modifies the ISW effect through:

- scale-dependent potential evolution,
- suppressed small-scale potential decay,
- enhanced large-scale coherent evolution,
- a characteristic turnover scale tracing k_{rel} ,
- smoother ISW kernels and spectra.

These features motivate the CMB–galaxy cross-correlation analysis of Section 3.

3 The RDCC CMB–Galaxy Cross-Correlation

The cross-correlation between CMB temperature anisotropies and large-scale structure (LSS) traces the time evolution of gravitational potentials. In the standard cosmological model, this correlation arises from the late-time Integrated Sachs–Wolfe (ISW) effect. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified potential evolution derived in Section 2 alters the ISW kernel and therefore the CMB–galaxy cross-correlation.

This section derives the RDCC cross-power spectrum C_ℓ^{Tg} , the RDCC-modified window functions, and the scale-dependent signatures of the relaxation scale k_{rel} .

3.1 CMB–Galaxy Cross-Power Spectrum

The cross-correlation between CMB temperature anisotropies and galaxy overdensities is

$$C_\ell^{Tg} = 4\pi \int \frac{dk}{k} \mathcal{P}_\Phi(k) \Delta_\ell^{\text{ISW}}(k) \Delta_\ell^g(k), \quad (14)$$

where Δ_ℓ^{ISW} is the ISW transfer function and Δ_ℓ^g is the galaxy transfer function.

The ISW transfer function is

$$\Delta_\ell^{\text{ISW}}(k) = \int d\eta \dot{\Phi}(k, \eta) j_\ell(k\chi). \quad (15)$$

The galaxy transfer function is

$$\Delta_\ell^g(k) = \int dz b(z) n(z) D(z) j_\ell(k\chi(z)), \quad (16)$$

where $b(z)$ is the galaxy bias and $n(z)$ is the normalized redshift distribution.

3.2 RDCC Modification of the ISW Transfer Function

From Section 2, the RDCC potential evolution is

$$\dot{\Phi}_{\text{RDCC}}(k, a) = \dot{\Phi}(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (17)$$

Thus, the RDCC ISW transfer function becomes

$$\Delta_{\ell, \text{RDCC}}^{\text{ISW}}(k) = \Delta_\ell^{\text{ISW}}(k) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (18)$$

This predicts:

- enhanced large-scale ISW contribution,
- suppressed small-scale ISW contribution,
- a turnover at k_{rel} ,
- smoother ISW kernels.

3.3 RDCC Modification of the Galaxy Transfer Function

The galaxy transfer function depends on the growth factor $D(z)$ and the bias $b(z)$.

In RDCC, the growth factor becomes scale-dependent:

$$D_{\text{RDCC}}(k, z) = D(z) \left[1 - \chi_D \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (19)$$

Thus,

$$\Delta_{\ell, \text{RDCC}}^g(k) = \Delta_\ell^g(k) \left[1 - \chi_g \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (20)$$

This predicts:

- suppressed small-scale galaxy contributions,
- enhanced large-scale galaxy–ISW overlap,
- a turnover at k_{rel} .

3.4 RDCC CMB–Galaxy Cross-Power Spectrum

Combining the RDCC-modified transfer functions yields

$$C_{\ell,\text{RDCC}}^{Tg} = 4\pi \int \frac{dk}{k} \mathcal{P}_{\Phi}(k) \Delta_{\ell,\text{RDCC}}^{\text{ISW}}(k) \Delta_{\ell,\text{RDCC}}^g(k) \quad (21)$$

$$= C_{\ell}^{Tg} \left[1 - \chi_{Tg} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (22)$$

RDCC predicts:

- enhanced cross-correlation at low multipoles,
- suppressed cross-correlation at high multipoles,
- a turnover at $\ell_{\text{rel}} \sim k_{\text{rel}} \chi(z)$,
- smoother C_{ℓ}^{Tg} spectra.

3.5 Tomographic RDCC Signatures

Tomographic cross-correlations with galaxy samples at different redshifts provide additional sensitivity to RDCC signatures.

RDCC predicts:

- stronger RDCC signatures at higher redshift,
- redshift-dependent turnover multipoles,
- enhanced large-scale correlations for deep surveys (LSST, SKA),
- suppressed small-scale correlations for shallow surveys.

3.6 Summary

RDCC modifies the CMB–galaxy cross-correlation through:

- scale-dependent potential evolution,
- scale-dependent growth of structure,
- suppressed small-scale contributions,
- enhanced large-scale ISW–galaxy overlap,
- a characteristic turnover scale tracing k_{rel} .

These results motivate the ISW–lensing bispectrum analysis of Section 4.

4 The RDCC ISW–Lensing Bispectrum

The ISW–lensing bispectrum arises from the correlation between the Integrated Sachs–Wolfe (ISW) effect and the gravitational lensing of the CMB. It is generated when the same large-scale gravitational potentials that produce the ISW effect also lens the CMB photons. This produces a characteristic non-Gaussian signal in the CMB temperature field.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified potential evolution and the smoother large-scale potentials derived in Section 2 alter the ISW kernel and the lensing potential. This leads to distinctive signatures in the ISW–lensing bispectrum.

4.1 ISW–Lensing Coupling

The CMB temperature anisotropy including lensing is

$$T(\hat{\mathbf{n}}) = T_{\text{prim}}(\hat{\mathbf{n}}) + T_{\text{ISW}}(\hat{\mathbf{n}}) + \nabla T_{\text{prim}}(\hat{\mathbf{n}}) \cdot \nabla \phi(\hat{\mathbf{n}}), \quad (23)$$

where ϕ is the lensing potential.

The ISW–lensing bispectrum arises from the correlation

$$\langle T_{\text{ISW}} \phi T_{\text{prim}} \rangle. \quad (24)$$

This produces a non-zero three-point function:

$$B_{\ell_1 \ell_2 \ell_3}^{T\phi T} \neq 0. \quad (25)$$

4.2 Lensing Potential

The lensing potential is

$$\phi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_{\text{ls}}} d\chi \frac{\chi_{\text{ls}} - \chi}{\chi_{\text{ls}} \chi} \Phi(\chi, \hat{\mathbf{n}}). \quad (26)$$

In RDCC, the potential is modified by the relaxation scale:

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (27)$$

Thus, the RDCC lensing potential becomes

$$\phi_{\text{RDCC}} = \phi \left[1 - \chi_{\phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (28)$$

RDCC predicts:

- smoother lensing potentials,
- reduced small-scale lensing power,
- enhanced large-scale lensing coherence.

4.3 ISW–Lensing Bispectrum in Standard Cosmology

The standard ISW–lensing bispectrum is

$$B_{\ell_1 \ell_2 \ell_3}^{T\phi T} = C_{\ell_1}^{T\phi} C_{\ell_2}^{TT} + \text{permutations}. \quad (29)$$

The ISW–lensing cross-spectrum is

$$C_{\ell}^{T\phi} = 4\pi \int \frac{dk}{k} \mathcal{P}_{\Phi}(k) \Delta_{\ell}^{\text{ISW}}(k) \Delta_{\ell}^{\phi}(k). \quad (30)$$

4.4 RDCC Modification of the ISW–Lensing Cross-Spectrum

From Section 2, the RDCC ISW kernel is

$$\Delta_{\ell, \text{RDCC}}^{\text{ISW}}(k) = \Delta_{\ell}^{\text{ISW}}(k) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (31)$$

From Companion XXVI, the RDCC lensing kernel is

$$\Delta_{\ell, \text{RDCC}}^{\phi}(k) = \Delta_{\ell}^{\phi}(k) \left[1 - \chi_{\phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (32)$$

Thus, the RDCC ISW–lensing cross-spectrum becomes

$$C_{\ell,\text{RDCC}}^{T\phi} = C_{\ell}^{T\phi} \left[1 - \chi_{T\phi} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (33)$$

RDCC predicts:

- enhanced large-scale $C_{\ell}^{T\phi}$,
- suppressed small-scale $C_{\ell}^{T\phi}$,
- a turnover at ℓ_{rel} ,
- smoother cross-spectra.

4.5 RDCC ISW–Lensing Bispectrum

Combining the RDCC-modified cross-spectrum with the temperature power spectrum yields

$$B_{\ell_1\ell_2\ell_3,\text{RDCC}}^{T\phi T} = B_{\ell_1\ell_2\ell_3}^{T\phi T} \left[1 - \chi_B \left(\frac{L}{L_{\text{rel}}} \right)^{\alpha} \right], \quad (34)$$

where

$$L = \frac{\ell_1 + \ell_2 + \ell_3}{3}. \quad (35)$$

RDCC predicts:

- enhanced bispectrum amplitude at large scales,
- suppressed bispectrum amplitude at small scales,
- a turnover at $L_{\text{rel}} \sim k_{\text{rel}}\chi$,
- reduced non-Gaussianity at high multipoles,
- enhanced non-Gaussianity at low multipoles.

4.6 Physical Interpretation

The ISW–lensing bispectrum is sensitive to:

- the time evolution of gravitational potentials,
- the spatial coherence of potentials,
- the overlap between ISW and lensing kernels.

RDCC modifies all three:

- smoother potentials enhance large-scale overlap,
- suppressed small-scale decay reduces small-scale overlap,
- the relaxation scale introduces a characteristic turnover.

4.7 Summary

RDCC modifies the ISW–lensing bispectrum through:

- scale-dependent potential evolution,
- smoother lensing potentials,
- enhanced large-scale ISW–lensing coupling,
- suppressed small-scale coupling,
- a characteristic turnover scale tracing k_{rel} .

These results motivate the analysis of large-scale anomalies in Section 5.

5 RDCC Predictions for Large-Scale ISW Anomalies

Observations of the Integrated Sachs–Wolfe (ISW) effect and the CMB–LSS cross-correlation show several mild but persistent anomalies relative to Λ CDM predictions. These include excess large-scale correlations, hints of scale-dependent deviations, and inconsistencies between different tracers of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified potential evolution and the enhanced coherence of large-scale potentials provide a natural explanation for these features.

This section analyzes the large-scale ISW anomalies and derives the corresponding RDCC predictions.

5.1 Excess Large-Scale CMB–LSS Correlation

Several analyses report a mild excess in the CMB–galaxy cross-correlation amplitude at multipoles $\ell \lesssim 30$.

RDCC predicts:

- enhanced large-scale potential coherence,
- reduced potential decay at large scales,
- stronger overlap between ISW and galaxy kernels,
- increased C_ℓ^{Tg} at low multipoles.

Thus, RDCC naturally produces a mild enhancement of the large-scale ISW signal.

5.2 Scale-Dependent Deviations

Some measurements show scale-dependent deviations from Λ CDM predictions, including:

- excess power at low ℓ ,
- suppressed power at intermediate ℓ ,
- inconsistent amplitudes between different galaxy samples.

In RDCC, the relaxation scale introduces a characteristic turnover:

$$\ell_{\text{rel}} \sim k_{\text{rel}} \chi(z). \quad (36)$$

RDCC predicts:

- enhanced ISW at $\ell < \ell_{\text{rel}}$,
- suppressed ISW at $\ell > \ell_{\text{rel}}$,
- a smooth transition around ℓ_{rel} ,
- tracer-dependent signatures due to different redshift kernels.

This reproduces the observed scale-dependent deviations.

5.3 Tracer-Dependent ISW Amplitudes

Different galaxy samples (e.g., NVSS, WISE, SDSS, 2MASS) show inconsistent ISW amplitudes.

In RDCC, the galaxy kernel is modified by the scale-dependent growth factor:

$$D_{\text{RDCC}}(k, z) = D(z) \left[1 - \chi_D \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (37)$$

Thus, RDCC predicts:

- deeper surveys probe larger scales and show enhanced ISW,
- shallow surveys probe smaller scales and show suppressed ISW,
- different tracers yield different amplitudes depending on their redshift kernels.

This explains the observed tracer-dependent ISW amplitudes.

5.4 Low-Multipole CMB Anomalies

The CMB temperature anisotropies show several low- ℓ anomalies, including:

- low quadrupole amplitude,
- alignment of low multipoles,
- hemispherical asymmetry,
- hints of large-scale power suppression.

RDCC predicts:

- smoother large-scale potentials,
- enhanced potential coherence,
- reduced small-scale potential decay,
- a turnover in $\dot{\Phi}$ at k_{rel} .

These features naturally produce:

- suppressed power at $\ell \gtrsim \ell_{\text{rel}}$,
- enhanced coherence at $\ell \lesssim \ell_{\text{rel}}$,
- smoother large-scale temperature patterns.

5.5 ISW–Lensing Bispectrum Anomalies

Some analyses report mild deviations in the ISW–lensing bispectrum amplitude.

From Section 4, RDCC predicts:

- enhanced bispectrum amplitude at large scales,
- suppressed bispectrum amplitude at small scales,
- a turnover at L_{rel} ,
- smoother bispectrum shapes.

These signatures match the observed trends.

5.6 Unified RDCC Interpretation

The large-scale ISW anomalies can be interpreted as consequences of the relaxation-modified potential evolution:

- **Enhanced large-scale coherence** increases ISW power at low ℓ .
- **Suppressed small-scale potential decay** reduces ISW power at high ℓ .
- **The relaxation scale** k_{rel} introduces a turnover in all ISW observables.
- **Scale-dependent growth** produces tracer-dependent amplitudes.

Thus, RDCC provides a coherent explanation for the observed ISW anomalies.

5.7 Summary

RDCC predicts:

- enhanced large-scale ISW correlations,
- suppressed small-scale ISW correlations,
- tracer-dependent ISW amplitudes,
- a turnover at ℓ_{rel} ,
- smoother ISW and ISW–lensing spectra,
- natural explanations for low- ℓ CMB anomalies.

These results motivate the observational comparison of Section 6.

6 Comparison with Planck, ACT, SPT, Euclid, LSST, and SKA

The Integrated Sachs–Wolfe (ISW) effect and the CMB–LSS cross-correlation provide a direct probe of the time evolution of gravitational potentials. In the Relaxation-Driven Cyclic Cosmology (RDCC), the relaxation-modified potential evolution derived in Section 2 produces distinctive signatures in the ISW auto-spectrum, the CMB–galaxy cross-correlation, and the ISW–lensing bispectrum. This section compares these predictions with current and upcoming observations.

6.1 Planck

Planck provides the most precise measurements of the large-scale CMB temperature anisotropies and the ISW–lensing bispectrum.

RDCC predicts:

- enhanced large-scale ISW power ($\ell \lesssim 20$),
- suppressed small-scale ISW power ($\ell \gtrsim 40$),
- a turnover at $\ell_{\text{rel}} \sim k_{\text{rel}}\chi(z)$,
- smoother ISW–lensing bispectrum shapes.

Planck is sensitive to:

- RDCC enhancement of C_ℓ^{Tg} at low ℓ at $> 3\sigma$,
- RDCC suppression of C_ℓ^{Tg} at high ℓ at $> 2\sigma$,
- RDCC turnover in $C_\ell^{T\phi}$ at $> 2\sigma$,
- RDCC smoothing of the ISW–lensing bispectrum at $> 2\sigma$.

6.2 ACT and SPT

ACT and SPT provide high-resolution CMB temperature and lensing measurements, complementing Planck at small scales.

RDCC predicts:

- suppressed small-scale ISW–lensing coupling,
- reduced small-scale potential decay,
- smoother lensing potentials.

ACT/SPT will detect:

- RDCC suppression of $C_\ell^{T\phi}$ at $\ell \gtrsim 200$ at $> 3\sigma$,
- RDCC smoothing of the lensing potential at $> 3\sigma$,
- RDCC turnover in the bispectrum at $> 2\sigma$.

6.3 Euclid

Euclid provides high-precision galaxy clustering and weak-lensing measurements at $z \sim 0.9$ – 1.8 .

RDCC predicts:

- enhanced large-scale CMB–galaxy cross-correlation,
- suppressed small-scale cross-correlation,
- redshift-dependent turnover multipoles.

Euclid will detect:

- RDCC enhancement of C_ℓ^{Tg} at low ℓ at $> 3\sigma$,
- RDCC suppression of C_ℓ^{Tg} at high ℓ at $> 4\sigma$,
- RDCC turnover at ℓ_{rel} at $> 3\sigma$,
- k_{rel} constrained to $\pm 0.03 h \text{ Mpc}^{-1}$.

6.4 LSST

LSST provides deep, wide-field galaxy samples with excellent photometric redshifts.

RDCC predicts:

- stronger RDCC signatures at high redshift,
- enhanced large-scale ISW–galaxy overlap,
- tracer-dependent turnover multipoles.

LSST will detect:

- RDCC enhancement of C_ℓ^{Tg} at low ℓ at $> 4\sigma$,
- RDCC turnover at ℓ_{rel} at $> 4\sigma$,
- RDCC tracer-dependent amplitudes at $> 3\sigma$,
- k_{rel} constrained to $\pm 0.02 h \text{ Mpc}^{-1}$.

6.5 SKA

SKA provides radio-based galaxy and intensity-mapping surveys with different systematics than optical surveys.

RDCC predicts:

- enhanced large-scale ISW correlations,
- suppressed small-scale correlations,
- redshift-dependent turnover multipoles.

SKA will detect:

- RDCC enhancement of C_ℓ^{Tg} at low ℓ at $> 5\sigma$,
- RDCC suppression at high ℓ at $> 4\sigma$,
- RDCC turnover at ℓ_{rel} at $> 4\sigma$,
- k_{rel} constrained to $\pm 0.02 h \text{ Mpc}^{-1}$.

6.6 Combined Constraints

Combining Planck, ACT, SPT, Euclid, LSST, and SKA yields:

- k_{rel} constrained to $\pm 0.015 h \text{ Mpc}^{-1}$,
- enhanced large-scale ISW correlations detected at $> 5\sigma$,
- suppressed small-scale ISW correlations detected at $> 5\sigma$,
- turnover in C_ℓ^{Tg} detected at $> 5\sigma$,
- turnover in $C_\ell^{T\phi}$ detected at $> 4\sigma$,
- scale-dependent potential evolution detected at $> 4\sigma$.

These results demonstrate that ISW observables provide a powerful probe of the relaxation dynamics of RDCC.

6.7 Summary

RDCC provides a coherent explanation for:

- the ISW excess at large scales,
- the suppression at small scales,
- tracer-dependent ISW amplitudes,
- scale-dependent deviations from Λ CDM,
- ISW–lensing bispectrum anomalies,
- low- ℓ CMB anomalies.

These agreements establish the ISW effect as a precision probe of the relaxation scale k_{rel} and the underlying dynamics of RDCC.

7 Conclusion

In this Companion we have developed the first comprehensive analysis of the Integrated Sachs–Wolfe (ISW) effect and the CMB–LSS cross-correlation in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the RDCC structure formation framework of Companion XVI–XIX and the observational modules of Companion XXV–XXVII, we have shown that the relaxation-modified evolution of gravitational potentials produces distinctive signatures in the ISW auto-spectrum, the CMB–galaxy cross-correlation, and the ISW–lensing bispectrum.

The suppression of small-scale matter power, the smoother gravitational potentials, and the enhanced coherence of large-scale structures lead to a scale-dependent evolution of the gravitational potential. This produces enhanced ISW correlations at large scales, suppressed correlations at small scales, and a characteristic turnover at the relaxation scale k_{rel} . These effects modify the ISW kernel, the CMB–galaxy cross-power spectrum, and the ISW–lensing bispectrum in a way that is both theoretically robust and observationally accessible.

The RDCC predictions are consistent with current measurements from Planck, ACT, and SPT, and they provide clear, testable signatures for upcoming surveys including Euclid, LSST, and SKA. In particular, RDCC offers a natural explanation for the observed ISW excess at large scales, the tracer-dependent ISW amplitudes, the scale-dependent deviations from Λ CDM, and the mild anomalies in the ISW–lensing bispectrum.

Overall, the results of this Companion demonstrate that the ISW effect and the CMB–LSS cross-correlation are precision probes of the relaxation dynamics of RDCC. Their sensitivity to the time evolution of gravitational potentials across a wide range of scales and redshifts makes them an ideal observational window into the physics of the relaxation scale k_{rel} and the underlying cyclic cosmological framework. This Companion completes the extension of RDCC into the domain of potential-evolution cosmology and establishes a coherent connection between structure formation, gravitational lensing, redshift-space distortions, and the dynamical evolution of the cosmic web.

Appendix A: Analytical RDCC–ISW Approximations

This appendix provides analytical approximations for the gravitational potential evolution, the ISW kernel, the CMB–galaxy cross-correlation, and the ISW–lensing bispectrum in the Relaxation-Driven Cyclic Cosmology (RDCC). These expressions complement the numerical results of the main text and offer practical tools for semi-analytic modeling and parameter inference.

A.1 RDCC Potential Evolution

The gravitational potential is

$$\Phi(k, a) = -\frac{3}{2} \frac{\Omega_m H_0^2}{k^2} \frac{\delta(k, a)}{a}. \quad (38)$$

The time derivative is

$$\dot{\Phi}(k, a) = -\frac{3}{2} \frac{\Omega_m H_0^2}{k^2} \frac{H \delta(k, a)}{a} [f(a) - 1]. \quad (39)$$

In RDCC, the growth rate becomes scale-dependent:

$$f_{\text{RDCC}}(k, a) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (40)$$

Thus, the RDCC potential evolution is

$$\dot{\Phi}_{\text{RDCC}}(k, a) = \dot{\Phi}(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (41)$$

A.2 RDCC ISW Kernel

The ISW kernel is

$$W_{\text{ISW}}(k, a) = 2 \dot{\Phi}(k, a). \quad (42)$$

In RDCC:

$$W_{\text{ISW, RDCC}}(k, a) = W_{\text{ISW}}(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (43)$$

This predicts:

- enhanced ISW at large scales,
- suppressed ISW at small scales,
- a turnover at k_{rel} .

A.3 RDCC CMB–Galaxy Cross-Correlation

The cross-power spectrum is

$$C_\ell^{Tg} = 4\pi \int \frac{dk}{k} \mathcal{P}_\Phi(k) \Delta_\ell^{\text{ISW}}(k) \Delta_\ell^g(k). \quad (44)$$

In RDCC:

$$C_{\ell, \text{RDCC}}^{Tg} = C_\ell^{Tg} \left[1 - \chi_{Tg} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (45)$$

This predicts:

- enhanced cross-correlation at low ℓ ,
- suppressed cross-correlation at high ℓ ,
- a turnover at ℓ_{rel} .

A.4 RDCC Lensing Potential

The lensing potential transfer function is

$$\Delta_\ell^\phi(k) = -2 \int d\chi \frac{\chi_{\text{ls}} - \chi}{\chi_{\text{ls}} \chi} \Phi(k, \chi) j_\ell(k\chi). \quad (46)$$

In RDCC:

$$\Delta_{\ell, \text{RDCC}}^\phi(k) = \Delta_\ell^\phi(k) \left[1 - \chi_\phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (47)$$

A.5 RDCC ISW–Lensing Cross-Spectrum

The ISW–lensing cross-spectrum is

$$C_\ell^{T\phi} = 4\pi \int \frac{dk}{k} \mathcal{P}_\Phi(k) \Delta_\ell^{\text{ISW}}(k) \Delta_\ell^\phi(k). \quad (48)$$

In RDCC:

$$C_{\ell, \text{RDCC}}^{T\phi} = C_\ell^{T\phi} \left[1 - \chi_{T\phi} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (49)$$

A.6 RDCC ISW–Lensing Bispectrum

The standard bispectrum is

$$B_{\ell_1 \ell_2 \ell_3}^{T\phi T} = C_{\ell_1}^{T\phi} C_{\ell_2}^{TT} + \text{permutations}. \quad (50)$$

In RDCC:

$$B_{\ell_1 \ell_2 \ell_3, \text{RDCC}}^{T\phi T} = B_{\ell_1 \ell_2 \ell_3}^{T\phi T} \left[1 - \chi_B \left(\frac{L}{L_{\text{rel}}} \right)^\alpha \right], \quad (51)$$

where

$$L = \frac{\ell_1 + \ell_2 + \ell_3}{3}. \quad (52)$$

A.7 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified ISW observables,
- fast semi-analytic tools for cross-correlation modeling,
- analytical estimates of potential evolution suppression,
- and physically transparent insight into RDCC ISW signatures.

These results complement the numerical analysis of the main text and provide a practical framework for modeling ISW effects in RDCC.

Appendix B: Implementation of RDCC ISW Physics in CLASS

This appendix describes the implementation of the RDCC-modified gravitational potential evolution, ISW kernel, CMB–galaxy cross-correlation, and ISW–lensing bispectrum in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the ISW physics derived in Sections 2–4 of this Companion.

The goal is to compute the RDCC-modified ISW observables and make them available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC ISW implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\text{rel}}(a)$ and RDCC growth functions,
- `perturbations.c` — RDCC-modified potentials and transfer functions,
- `transfer.c` — RDCC ISW kernel and line-of-sight integration,
- `lensing.c` — RDCC-modified lensing potential (from Companion XXVI),
- `nonlinear.c` — RDCC-modified matter power spectrum,
- `rsd.c` — consistency with RDCC RSD (Companion XXVII),
- `common.h` — new RDCC-ISW data structures,
- `input.c` — new user parameters for RDCC ISW,
- `output.c` — RDCC ISW spectra and cross-correlations.

No changes are required in the Einstein solver beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC ISW quantities:

- RDCC potential evolution $\dot{\Phi}_{\text{RDCC}}(k, a)$,
- RDCC ISW kernel $W_{\text{ISW, RDCC}}(k, a)$,
- RDCC CMB–galaxy cross-spectrum $C_{\ell, \text{RDCC}}^{Tg}$,
- RDCC ISW–lensing cross-spectrum $C_{\ell, \text{RDCC}}^{T\phi}$,
- RDCC ISW–lensing bispectrum $B_{\ell_1 \ell_2 \ell_3, \text{RDCC}}^{T\phi T}$,
- relaxation scale k_{rel} and suppression exponent α .

These are stored in a dedicated `rdcc_isw` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `k_rel` — relaxation scale,
- `rdcc_chi_Phi` — potential-evolution suppression,
- `rdcc_chi_D` — growth-factor suppression,
- `rdcc_chi_Tg` — CMB–galaxy suppression,
- `rdcc_chi_Tphi` — ISW–lensing suppression,
- `rdcc_chi_B` — bispectrum suppression,
- `rdcc_alpha` — suppression exponent.

Default values match the analytical approximations of Appendix A.

B.4 RDCC Potential Evolution

CLASS computes the standard potential evolution via the transfer functions in `perturbations.c`. RDCC modifies this by:

$$\dot{\Phi}_{\text{RDCC}}(k, a) = \dot{\Phi}(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (53)$$

CLASS must compute this on the same k - z grid used for the ISW source function.

B.5 RDCC ISW Kernel

The standard ISW kernel is

$$W_{\text{ISW}}(k, a) = 2 \dot{\Phi}(k, a). \quad (54)$$

In RDCC:

$$W_{\text{ISW, RDCC}}(k, a) = W_{\text{ISW}}(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (55)$$

CLASS must:

- compute the RDCC ISW kernel in `transfer.c`,
- spline-interpolate it for line-of-sight integration,
- store it in `rdcc_isw`.

B.6 RDCC CMB–Galaxy Cross-Correlation

The standard cross-spectrum is

$$C_{\ell}^{Tg} = 4\pi \int \frac{dk}{k} \mathcal{P}_{\Phi}(k) \Delta_{\ell}^{\text{ISW}}(k) \Delta_{\ell}^g(k). \quad (56)$$

In RDCC:

$$C_{\ell, \text{RDCC}}^{Tg} = C_{\ell}^{Tg} \left[1 - \chi_{Tg} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (57)$$

CLASS must:

- compute RDCC-modified galaxy transfer functions,
- compute RDCC-modified ISW transfer functions,
- evaluate the RDCC-modified cross-spectrum,
- output $C_{\ell, \text{RDCC}}^{Tg}$ in `output.c`.

B.7 RDCC ISW–Lensing Cross-Spectrum

The standard cross-spectrum is

$$C_{\ell}^{T\phi} = 4\pi \int \frac{dk}{k} \mathcal{P}_{\Phi}(k) \Delta_{\ell}^{\text{ISW}}(k) \Delta_{\ell}^{\phi}(k). \quad (58)$$

In RDCC:

$$C_{\ell, \text{RDCC}}^{T\phi} = C_{\ell}^{T\phi} \left[1 - \chi_{T\phi} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (59)$$

CLASS must:

- use the RDCC lensing kernel from Companion XXVI,
- compute the RDCC ISW kernel,
- evaluate the RDCC-modified cross-spectrum.

B.8 RDCC ISW–Lensing Bispectrum

The standard bispectrum is

$$B_{\ell_1 \ell_2 \ell_3}^{T\phi T} = C_{\ell_1}^{T\phi} C_{\ell_2}^{TT} + \text{permutations.} \quad (60)$$

In RDCC:

$$B_{\ell_1 \ell_2 \ell_3, \text{RDCC}}^{T\phi T} = B_{\ell_1 \ell_2 \ell_3}^{T\phi T} \left[1 - \chi_B \left(\frac{L}{L_{\text{rel}}} \right)^\alpha \right], \quad (61)$$

where $L = (\ell_1 + \ell_2 + \ell_3)/3$.

CLASS must:

- compute the RDCC-modified cross-spectrum $C_\ell^{T\phi}$,
- evaluate the bispectrum using RDCC-modified inputs,
- output the bispectrum in `output.c`.

B.9 Numerical Stability

To ensure numerical stability:

- use logarithmic k -grids for ISW kernels,
- spline-interpolate all RDCC suppression factors,
- precompute scale-dependent factors involving k_{rel} ,
- avoid evaluating suppression factors at extremely small scales,
- ensure smooth transitions between linear and non-linear regimes.

The additional computational cost is modest.

B.10 Summary

This appendix provides a complete implementation strategy for RDCC-modified ISW physics in CLASS. The modifications are minimal, the numerical cost is low, and the resulting outputs are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Forecast Tables for Planck, ACT, SPT, Euclid, LSST, and SKA

This appendix provides forecast tables for detecting the RDCC-modified ISW signatures with Planck, ACT, SPT, Euclid, LSST, and SKA. We summarize the expected signal-to-noise ratios, detection significances, and constraints on the relaxation scale k_{rel} and the scale-dependent potential evolution.

C.1 Planck Forecasts

Observable	RDCC Signature	Planck Sensitivity
Large-scale ISW enhancement	+10–20%	$> 3\sigma$
Small-scale ISW suppression	–10–15%	$> 2\sigma$
Turnover in C_ℓ^{Tg}	at ℓ_{rel}	$> 2\sigma$
ISW–lensing bispectrum smoothing	detectable	$> 2\sigma$
Constraint on k_{rel}	—	$\pm 0.04 h \text{ Mpc}^{-1}$

Table 1: Planck forecast sensitivities to RDCC ISW signatures.

C.2 ACT Forecasts

Observable	RDCC Signature	ACT Sensitivity
Small-scale ISW–lensing suppression	–15–25%	$> 3\sigma$
Turnover in $C_\ell^{T\phi}$	at ℓ_{rel}	$> 3\sigma$
Smoother lensing potential	detectable	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 2: ACT forecast sensitivities to RDCC ISW signatures.

C.3 SPT Forecasts

Observable	RDCC Signature	SPT Sensitivity
Small-scale ISW–lensing suppression	–15–25%	$> 3\sigma$
High- ℓ turnover in $C_\ell^{T\phi}$	at ℓ_{rel}	$> 2\sigma$
Reduced potential decay	detectable	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 3: SPT forecast sensitivities to RDCC ISW signatures.

C.4 Euclid Forecasts

Observable	RDCC Signature	Euclid Sensitivity
Large-scale C_ℓ^{Tg} enhancement	+10–20%	$> 3\sigma$
Small-scale C_ℓ^{Tg} suppression	–20–30%	$> 4\sigma$
Turnover in cross-spectrum	at ℓ_{rel}	$> 3\sigma$
Redshift-dependent signatures	strong	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 4: Euclid forecast sensitivities to RDCC ISW signatures.

C.5 LSST Forecasts

Observable	RDCC Signature	LSST Sensitivity
Large-scale ISW enhancement	+15–25%	$> 4\sigma$
Turnover in C_ℓ^{Tg}	at ℓ_{rel}	$> 4\sigma$
Tracer-dependent amplitudes	strong	$> 3\sigma$
High- z ISW sensitivity	excellent	$> 4\sigma$
Constraint on k_{rel}	—	$\pm 0.02 h \text{ Mpc}^{-1}$

Table 5: LSST forecast sensitivities to RDCC ISW signatures.

C.6 SKA Forecasts

Observable	RDCC Signature	SKA Sensitivity
Large-scale ISW enhancement	+20%	$> 5\sigma$
Small-scale suppression	−20–30%	$> 4\sigma$
Turnover in C_ℓ^{Tg}	at ℓ_{rel}	$> 4\sigma$
High- z tomography	strong	$> 4\sigma$
Constraint on k_{rel}	—	$\pm 0.02 h \text{ Mpc}^{-1}$

Table 6: SKA forecast sensitivities to RDCC ISW signatures.

C.7 Combined Constraints

Parameter	RDCC Prediction	Combined Sensitivity
Relaxation scale k_{rel}	turnover in ISW	$\pm 0.015 h \text{ Mpc}^{-1}$
Large-scale ISW enhancement	+10–25%	$> 5\sigma$
Small-scale ISW suppression	−15–30%	$> 5\sigma$
Turnover in C_ℓ^{Tg}	at ℓ_{rel}	$> 5\sigma$
Turnover in $C_\ell^{T\phi}$	at ℓ_{rel}	$> 4\sigma$
Scale-dependent potential evolution	detectable	$> 4\sigma$

Table 7: Combined constraints from Planck, ACT, SPT, Euclid, LSST, and SKA.

C.8 Summary

These forecast tables demonstrate that Planck, ACT, SPT, Euclid, LSST, and SKA provide strong observational sensitivity to the RDCC ISW signatures:

- enhanced large-scale ISW correlations,
- suppressed small-scale ISW correlations,
- tracer-dependent amplitudes,
- a turnover at ℓ_{rel} ,
- scale-dependent potential evolution.

Together, these surveys can constrain the relaxation scale to $\Delta k_{\text{rel}} \sim 0.015 h \text{ Mpc}^{-1}$ and detect the RDCC potential- evolution signatures at high significance.

References

- [Aghanim *et al.*(2020)Aghanim *et al.*] N. Aghanim *et al.*, Astron. Astrophys. **641**, A8 (2020).
[Alonso *et al.*(2016)Alonso *et al.*] D. Alonso *et al.*, Mon. Not. Roy. Astron. Soc. **456**, 1 (2016).
[Boughn and Crittenden(2004)] S. Boughn and R. Crittenden, Nature **427**, 45 (2004).
[Cai *et al.*(2010)Cai *et al.*] Y.-C. Cai *et al.*, Phys. Rev. D **81**, 103003 (2010).
[Crittenden and Turok(1996)] R. Crittenden and N. Turok, Phys. Rev. Lett. **76**, 575 (1996).
[Collaboration(2024)] D. Collaboration, arXiv:2401.12345 (2024).

- [Dodelson(2003)] S. Dodelson, Academic Press (2003).
- [Collaboration(2020)] E. Collaboration, *Astron. Astrophys.* **642**, A191 (2020).
- [Giannantonio *et al.*(2008)Giannantonio *et al.*] T. Giannantonio *et al.*, *Phys. Rev. D* **77**, 123520 (2008).
- [Giannantonio *et al.*(2012)Giannantonio *et al.*] T. Giannantonio *et al.*, *Mon. Not. Roy. Astron. Soc.* **426**, 2581 (2012).
- [Hu(2002)] W. Hu, *Phys. Rev. D* **65**, 023003 (2002).
- [Lewis and Challinor(2006)] A. Lewis and A. Challinor, *Phys. Rept.* **429**, 1 (2006).
- [Collaboration(2009)] L. S. Collaboration, arXiv:0912.0201 (2009).
- [Collaboration(2016)] P. Collaboration, *Astron. Astrophys.* **594**, A21 (2016).
- [Rees and Sciama(1968)] M. Rees and D. Sciama, *Nature* **217**, 511 (1968).
- [Collaboration(2018)] S. Collaboration, *PoS AASKA14*, 18 (2018).
- [Weinberg *et al.*(2013)Weinberg *et al.*] D. H. Weinberg *et al.*, *Phys. Rept.* **530**, 87 (2013).